Neural Bellman Operators
Author Response to the Advisory Referee

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1 Report identity, scientific baseline and present author revision

This is a new author revision of *Neural Bellman Operators*, retaining the original recursive economic decision problem and the Econometrica numerical-methods objective. The completed author baseline is R120 source `39a20bf1fe011175857811bf67bacd788ae80c52`, with its source/PDF publication `cc58ec9613044f97aa1ae5c8d60ec1c3c0021dcd` and final delivery `f06f220aea525088e82228e8e9be4947fb81b1ca`. The R119 and R110 scientific ancestors, their original sources, negative results and incomplete independent episodes remain preserved. A larger revision label is not substituted for a completed source and its evidence.

The fresh 20 October 2026 observation found 361 remote branches and 303 distinct head commits, with no new persisted report or later completed author source. All 360 earlier branch heads are unchanged; the R120 branch has its verified final delivery. The read-only full-tree census enumerated 24,121,986 entries and 67,440 distinct blob identities across those heads. Its compact indexes preserve every commit, tree and stream hash. This is a complete file-identity census, not a claim that every historical program was rerun or every archived proof was freshly revalidated.

A subsequent complete remote-head observation, completed at 18:58:57 UTC on 20 October, found 362 branches. All 361 earlier heads were unchanged. The only addition was the present isolated R121 branch, then at its state-publication commit D. The observed heads and exact difference are retained in `provenance121/REMOTE_BRANCHES121_POST_CAMPAIGN.json` and `provenance121/POST_CAMPAIGN_BRANCH_DELTA121.json`. This refresh confirms the earlier controlling manuscript and report identities; it does not repeat the historical scientific experiments.

The controlling advisory remains report commit `9723c6cd2cc8b3e071422f3e522d4a843946db63`, report blob `33ba8b144e2f7816b19b036404341402d23a582f`, assessing manuscript `f802fe64d91d9f5c1c8f86123b0fbe114e622279`. Its SHA-256 is `479b28c8800084846336cac078877a01dbbd2bae546d179633a2734401da19b9`. It is the same owner-commissioned AI-assisted advisory, not an Econometric Society decision and not a review of R119, R120 or the present revision. All 35 distinct original report bodies, totaling 975,366 bytes, are preserved unchanged under `provenance121/reports121/`. The 36 review references include one reference without a distinct new report. The exact report and author-route custody index is `provenance121/reports121/REPORT_SOURCE_INDEX121.json`.

An author response rather than a change of research program. The literal redirection in M1 is not followed: the author expressly requests continued work on NBO and its original economic objectives at Econometrica. We address its scientific substance by centering the article on one decision problem and one testable mechanism. We do not count a venue change, a different

objective, or abandonment of neural operator research as completion. Representation-neutral mathematics is stated as such, and neural-specific economic value remains a question for equal-information and full-cost evidence.

Completed R120 evidence and the unresolved learning mechanism. R120 executed 4,096 primary and 6,144 descriptive timing decisions under its separate source, actual-model and new-state barriers. Independent original-primitive validation covers all 861 policy keys and 4,096 comparisons, including all-action optimum supports. Its fixed 48-group, 192-event inference family gives 42 positive learned-incumbent event-staircase mean-gain lower bounds, with its stated simultaneous 95% coverage. These are R120 findings.

They also identify the remaining problem. The new paired service differs from inherited refinement at only 12 of 1,024 actions, establishes no strict incremental cell-mean improvement and is slower in all 48 matched clocks. Direct original-model refinement is faster than the paired service in all those clocks. All 512 ReLU–hinge action pairs agree, and every learned completed cell has a strictly worse sample mean original objective than the online control. All 32 three-clock ReLU/full-menu upper-region intersections are empty; 12 individual-clock and two actual-primary upper regions remain nonempty and unresolved. The intersection result does not mean impossibility under every clock. None of these adverse findings is deleted or relabeled as a new R121 result.

New mathematical and algorithmic response. R121 changes acquisition and fitting in the same original economy. The root critic retains the known current action cost and learns the marginal of the integrated original continuation risk in original action units. Theorem 6 derives explicit regularity through binding capacity and constrained adverse-tail cuts. Proposition 7 converts paid original-risk value enclosures into outward marginal labels, including inward boundary probes. Proposition 8 gives exact primal and dual bounds for the actual bounded minimax readout. Theorem 9 transports those labels off design through finite feature ranges, including derivative jumps, and integrates their errors along feasible action paths. Proposition 10 connects an outward stationarity interval to original regret and action distance. The deployed-future allowance is paid once; the recursive value level is not discarded merely because an anchor cancels in a single action comparison.

The development explicitly credits the earlier action-null quotient, convex bounded-readout, constrained-stationarity, active-tail and paired comparison results. The new construction is assessed through its original integrated-risk regularity, actual paid labels, fitted bounded problem, off-design certificate and executed comparison. It is not a claim that quotients, spline or hinge geometry, convex readouts or the basic KKT inequality were invented in this revision.

Frozen numerical scope and completed execution. The scientific source and protocols were published at F, `16bf21334efcbc719b83e2ccf10e8527f8b956f2`, before any new original-model acquisition. Its native tree is `2666e47a9b5376904bf53ad187b97a850107fc30`; the exact manifest is `SOURCE_FREEZE121.json`. All 56 stored actors and both independent finite-fitting audits were then published at M, `796c5f2badf68b051ac6c144884c0e61b87faf5e`, before held-out entropy existed. D, `68eae434c0e61fb1a61c3271c651c83c28619a07`, contains the later 640-byte draw and 64 exact states, published before policy outcomes. The original continuous innovation law and economic objective are unchanged. The current mathematical sources are `new121/marginal_main.tex` and `new121/marginal_proofs.tex`; the executed study follows `PROTOCOL121.md`, `INDEPENDENT_PROTOCOL121.md` and `REPORTING_SPEC121.md`.

All 3,712 primary decisions and 9,280 decisions in five serial timing processes completed; every saved timing action matches its primary identity. The production records were published at R, `f6392f6909d280ecd310f152caab0fafe9091df7`, with 761 declared native Git blobs checked by `provenance121/R_PUBLICATION121.json`. Independent original-policy evaluation subsequently completed all 1,301 distinct policy/proposal keys, 167,829 terminal contexts and 1,406,549 assertions. It covers all delivered policies and all first proposals, including rejected ones. Independent evaluation started after the complete production/timing campaign while local HEAD was D and before R transport; it was not already running when D was originally published. The exact receipt correction and its prior bytes are preserved in the provenance.

The fixed statistical calculation and the deterministic reporter have also completed successfully. The reporter checks 473 bound inputs and 220,310 record assertions and reproduces 49 stable output files without new states, fitting, original economic evaluations or repeated clocks. The executed results are in `new121/results121.tex` and `new121/development121_results.tex`; exact rows, ledgers and successful stage receipts are under `results121/`.

What the completed study establishes and leaves open. All 28 MARGINAL treatments lower the finite-node marginal residual, but every independently enclosed 64-state sample mean gain $J_{\text{VALUE}} - J_{\text{MARGINAL}}$ is strictly negative. All 28 fixed joint population mean-gain intervals contain zero. Of 1,792 paired final actions, 889 differ. First rejections rise from 60 under VALUE to 294 under MARGINAL; all 3,712 delivered policies are qualified, with zero numerical service failures and zero exact fallbacks. Thus the safeguard works at its declared tolerance while the proposed learning objective fails to improve the original sample mean. Finite fitting was independently audited; the global fitted-feature cover needed for the stronger off-design theorem was not numerically certified.

MARGINAL complete primary cohort service is faster in four of 28 comparisons and slower in 24. Across five timing processes it is faster in 40 of 140 matched cohorts and slower in 100; including the allocated acquisition and required fit-audit bill gives 38 faster and 102 slower observations. Every primary ReLU–hinge raw and final action pair agrees. All ReLU full-menu sufficient and upper not-ruled-out price regions are empty in the 32 primary, 160 individual-clock and 32 all-five-clock comparisons. These are actual R121 findings, with their original state and clock scopes. They neither replace the distinct R120 evidence nor establish a population harm, neural-exclusive advantage or universal adoption impossibility.

The 64 fresh states are the statistical units; repeated methods, risks, hinges and clocks do not enlarge that sample. The study keeps its original model and Econometrica objective. Source/PDF identity and source-specific CI/publication are recorded separately in `SOURCE_READY121.json`, `EVIDENCE_MANIFEST121.json` and the subsequent actual publication receipts. Successful scientific execution and an assembly workflow are not substitutes for those distinct source-publication records.

2 Point-by-point response to the controlling advisory

The following preserves all 30 original requests exactly as quoted in the verified prior author response. Each item distinguishes the completed R120 action, the new R121 action and its continuing scientific boundary. The complete original text, line ranges, report identities, prior author treatment and current routes are retained in `provenance121/R97_REQUIREMENTS121.json`. A planned action and a mathematical subclaim are not blanket closure of the broader request. The current executed observations are integrated item by item, including adverse and statistically unresolved findings.

2.1 B1. Neural specificity remains unestablished

Original advisory request.

> Validity, recursive qualification, the AVaR enclosure, the information lower bound, and the fallback are representation-neutral. The ReLU family is not the complete-work winner and is never certified adopted over quadratic least squares.

Completed R120 author response. The completed study gives every representation the same four services and original-model information. The ordinary hinge receives the ReLU features and coefficients and pays the full learned-feature acquisition charge. All 512 ReLU–hinge action pairs are exactly equal. All 42 learned own-incumbent cells have strictly positive independent mean-gain lower bounds, but every one of the 14 learned completed cells has a strictly worse sample mean original objective than the online semi-analytic control. The paired service P is slower than inherited L in all 48 matched clocks. Every one of the 32 declared ReLU/full-conventional-menu price intersections across all three clocks is empty, including the intersections of upper not-ruled-out conditions. Thus the achieved contribution is an action-relevant Bellman comparison and original-policy correction; this equal-feature experiment supplies no neural-specific policy or adoption advantage.

The completed evidence for this item is in Table F.I, Table F.II, Table I.VIII; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The executed study now compares VALUE and MARGINAL within identical paid information, features, coefficient bounds and qualification services, and compares those actors with the full conventional menu. All 512 primary ReLU–hinge raw-proposal pairs and all 512 delivered-action pairs are equal; the hinge pays the ReLU acquisition and required audit costs. All 32 primary ReLU/full-menu sufficient and upper not-ruled-out price regions are empty, as are all 160 individual timing regions and all 32 five-clock intersections. Changing the learning objective therefore supplies no neural-specific policy or adoption advantage here. The marginal regularity, finite fitting and action-path results apply to every eligible representation and are stated as shared mathematics.

Remaining scientific requirement. A neural-specific contribution still requires a strict economic or information benefit against methods with the same features, information, feasible policies and full resource bill. The present empty recorded-menu regions do not prove impossibility for every economy, network, or deployment criterion.

Current execution and source route. Table II, Table B.VII, and the exact equal-feature rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.2 B2. The information-complexity result is too narrow for the platform claim

Original advisory request.

> The theorem is a specialized deterministic one-dimensional oracle result. It does not establish randomized, multivariate, nonuniform-law, or end-to-end Bellman complexity lower bounds.

Completed R120 author response. The new constrained-AVaR theorem supplies the original continuation gradient modulus and the action-distance modulus E. The fixed-known-tail information result has matching width order

$$\min\{KR(1-\alpha)N^{-2},\ ER(1-\alpha)N^{-1}\}.$$

One exact scalar value is one query; the claim is deterministic and univariate. The original smooth model also gives the stronger separate-width coefficient $2M_\alpha R^2(1-\alpha)^2$. The nonsmooth lower-bound adversary therefore does not establish an original-primitives or smooth-class lower bound. Full policy comparisons pay both risk deficits, outward point uncertainty and the unchanged future allowance. In execution the mixed upper is necessary at two recorded accepted levels for one common Poly3 candidate at $m = 4$; no wall-time gain or optimal general Bellman information complexity is inferred from that event. The full positive-margin refinement and future-information-floor propositions state when the interface can terminate successfully.

The completed evidence for this item is in Table I.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The new constructive upper result obtains outward action-marginal labels from integrated original-risk value enclosures, with explicit finite-difference, risk-level, capacity and rounding allowances. The two actually acquired budgets contain 650 and 2,106 original root-Q contexts across the two risks, respectively, rather than free derivative labels; 1,413,828 terminal label contexts are charged in total. VALUE receives the same complete bundle as MARGINAL. These executed information counts make the interface cost visible. They do not enlarge the class covered by the inherited deterministic univariate fixed-tail lower bound. The original-formula online method remains eligible and its actual cost is retained.

Remaining scientific requirement. No randomized, multivariate, nonuniform-law, formula-oracle, bit-complexity or end-to-end Bellman minimax lower bound is inferred from the earlier scalar oracle construction. Matching lower and upper bounds for the smooth original economy remain open at a separately specified information interface.

Current execution and source route. Proposition 7, Table B.III, and the acquisition rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.3 B3. The representation result is not decision-theoretic

Original advisory request.

> The ridge example separates restricted approximation classes but does not separate policies, service work, or economic outcomes. A conventional method with the same hinge is equivalent.

Completed R120 author response. The new comparison authorizes changes of the original investment policy. It treats a difference of convex continuations as Lipschitz, without assuming that the difference is convex. The independent original-policy and all-action-optimum witnesses now provide value, gain, regret and action-distance bounds for every returned decision. For ReLU

at $\alpha = 1/2$, P's mean original-policy gain over its own completed incumbent is approximately $[212.675, 213.885] \times 10^{-6}$; the mean-regret upper bound falls from 215.906×10^{-6} to 2.626×10^{-6}. The RMS action-error upper bound falls from 0.01470 to 0.001621. L returns the same ReLU policies, and conventional hinge is identical as well. These are original-investment improvements with a shared correction; they neither establish a neural geometry advantage nor borrow economic improvements from the separate posted-price example.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The new theorem controls the integral of continuation-marginal error along feasible action paths, treats feature derivative jumps explicitly, and keeps the state anchor when composing recursive values. The stationarity proposition converts an outward original-objective KKT interval into regret and action-distance bounds. The numerical distinction is now decisive: the finite-node marginal residual falls in all 28 paired treatments, but the independently enclosed 64-state mean gain $J_{\text{VALUE}} - J_{\text{MARGINAL}}$ is strictly negative in every treatment; 889 of 1,792 final action pairs differ. Thus fitting the reported node criterion better does not establish better decisions. The independent tables report original objectives, regret and action-error upper bounds, not only representation error.

Remaining scientific requirement. The off-design feature-range cover is a proved conditional construction but was not numerically certified for the fitted actors. Neither the node residual nor the finite optimizer gap can be substituted for that missing global premise, and no neural geometry separation follows from the shared hinge representation.

Current execution and source route. Theorem 9, Proposition 10, Table II, and Table B.I. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.4 B4. The certificate-aware objective fails its intended service connection

Original advisory request.

> It improves residual span in 36 of 36 cells but worsens aggregate rejections and most quadratic cells. The missing theorem is the connection from the trained objective to action-relevant resolution.

Completed R120 author response. The acquired information now targets the original continuation difference at two feasible actions. A paid common-tail mismatch and a strict complete-policy gate connect this information to a permissible decision change, with the true future-controller allowance included once. Holding the three inherited support-gradient proposal rules fixed, P changes 391 of 1,024 actions and accepts 815 steps; L changes 388 and accepts 807. Their final actions differ in only 12 cases, and none of the 16 cell-mean comparisons resolves a strict incremental P advantage. The same paid pair primitive is supplied to the stronger direct conventional proposal. All fitted critics still use the ordinary original-V_1 value-label objective. We therefore complete an action-dependent certificate and its original-policy evaluation, while retaining action-aware training and acquisition as a distinct unresolved scientific requirement.

The completed evidence for this item is in Table F.I, Table I.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. We implemented and independently audited the replacement objective, rather than defending the previous internal span. MARGINAL minimizes a bounded finite-node original-action risk-marginal error; its 28 exact primal/dual gap upper bounds lie between approximately 1.13×10^{-9} and 2.58×10^{-8}. All 28 node residuals improve, yet first rejections increase from 60 under VALUE to 294 under MARGINAL, all 28 independently enclosed sample mean gains are negative, and complete primary cohort service is slower in 24 of 28 comparisons. The fixed joint population intervals all contain zero. This is new adverse evidence about the proposed learning-to-service connection, not a positive result borrowed from R120.

Remaining scientific requirement. The intended economic improvement is not established by this experiment. A useful next algorithm must control the off-design action paths and qualification burden, with its acquisition and certification costs paid; the present finite training certificate does not certify that improvement or a positive population gain.

Current execution and source route. Proposition 8, Table II, Table B.II, and Table B.IV. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.5 B5. Complete work favors a strong conventional method

Original advisory request.

> Quadratic least squares is faster, uses fewer proposal queries, and has no neural acquisition overhead. R97 therefore does not shift the computational frontier toward NBO-specific representation.

Completed R120 author response. The completed four-service/eight-arm comparison retains all 4,096 primary and 6,144 timing decisions and charges actual common-label acquisition, each standalone fit and audit, full hinge acquisition, loading, proposals, tail witnesses, precision ladders, rejected work, fallback and cohort bookkeeping. Summing complete standalone 64-state costs over the same 16 cells gives 94.478 seconds for L, 132.729 for P and 80.302 for direct refinement D. P purchases fewer compiled terminal rows than L, yet is slower in every primary cell and all 48 matched fresh clocks. D purchases substantially more terminal rows and is faster than P in all those clocks. The independent rational study costs a separate 1,806.153 seconds and remains charged as research validation. These complete scopes replace a point-count-only competitiveness claim; they retain the stronger traditional method and every adverse clock.

The completed evidence for this item is in Table F.II; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The complete comparison includes quadratic, degree-two and degree-three polynomial, spline, grid, equal-feature hinge and direct original-formula online controls. Each learned pair pays for the same acquired label bundle

and its own fit and required rational audit; the hinge also pays its parent ReLU charge. All rejected proposals, conventional fallback, serialization and complete service clocks remain charged. The exact frontiers contain 1,392 volume-cost rows and 51,072 pairwise price constraints, including four prespecified reuse volumes. No ReLU full-menu sufficient region or upper not-ruled-out region survives in any primary or individual timing scenario. The conventional alternatives therefore remain essential to assessing whether learning buys an economic service.

Remaining scientific requirement. A finite recorded-menu cost frontier is not a universal computational lower bound. Acquisition allocations are derived from measured process components, not separate observed standalone fits, and the independent all-policy research validation is also reported in the study ledger even when it is not a deployment charge.

Current execution and source route. Table B.IV, Table B.V, Table B.VII, and the complete research ledger. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.6 B6. The purchase exercise is conditional sensitivity, not economic identification

Original advisory request.

> No learned service is adopted over quadratic least squares. Prices and fees are external, clocks are descriptive, and the policy interval is broad. The exercise does not change an economic conclusion.

Completed R120 author response. The fixed 48-group, 192-event population calculation has actually completed on the prospectively drawn 64 states. Under its stated finite-uniform population and independent-entropy assumption, the simultaneous 95% family gives positive event-staircase mean-gain lower bounds in all 42 learned own-incumbent groups, ranging from 2.661×10^{-6} to 14.192×10^{-6}. The six online groups have zero observed gains and zero lower bounds; the ordinary bounded-mean lower bounds alone are zero. The reported price sets use actual paid work and the declared conventional menu. Positive own-incumbent price regions coexist with empty ReLU/full-menu regions. No calibrated monetary value, preference parameter, market demand or population adoption conclusion is supplied by those normalized-objective calculations.

The completed evidence for this item is in Table F.III, Table I.VIII; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The completed economic comparison stays with the same original constrained investment objective and continuous innovation law. Reported prices value a second of computation in normalized objective units and solve the stated strict gain-minus-price-times-extra-cost inequalities exactly. The four reuse volumes and the full conventional menu were fixed before outcomes. The resulting 32 primary, 160 individual-clock and 32 all-five-clock ReLU comparisons have empty sufficient and upper not-ruled-out regions. Their meaning is a conditional decision-service comparison on the specified state cohorts, with exact endpoint and strictness flags, rather than an externally identified willingness to pay.

Remaining scientific requirement. The exercise does not identify welfare prices, an empirical adoption rate or a calibrated firm-level technology choice. Upper not-ruled-out intersections are only pairwise feasibility conditions, not jointly attainable adoption claims. A consequential calibrated application still requires its own data, information model and validation.

Current execution and source route. Table B.VII and the exact full-menu price constraints and menus. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.7 B7. The new 84 actors lack direct population policy-cost inference

Original advisory request.

> R97 transfers only the uniform deterministic qualification bound to the new services. The R96 statistical intervals belong to different frozen actors and cannot establish the new actors' mean-cost ordering.

Completed R120 author response. The source F, actual actors M and single 640-byte state draw D are separately and actually committed before evaluation. The recorded 64 exact 40-bit dyadic states are shared across all arms and risks, and singleton initial-state services prevent a batch fallback from making one row's rule depend on another row. All 4,096 original-policy comparisons and all 48-group/192-event statistics completed under these commitments. The 95% guarantee uses the fixed one-sided binomial allocation and bounded-mean allocation from the frozen protocol; it conditions on the declared independent uniform entropy law and fixed actors. This supplies the new R120 population evidence. It neither completes an old R97 actor family nor imports R98/R115 actors, reserved states or inference into this experiment.

The completed evidence for this item is in Table F.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The current 56 stored actors and both independent fit audits were published at M before the new 640-byte entropy draw; D then fixed 64 exact two-dimensional states before policy outcomes. Independent original-primitive evaluation covers all 1,301 distinct final-policy and raw-proposal keys and all 3,712 policy comparisons. The prespecified simultaneous family contains two optimum means, 58 regret means, 28 signed learning gains, 112 signed threshold events and 116 one-sided rejection endpoints, with joint coverage at least 95% under its stated sampling assumptions. All 28 signed population mean-gain intervals contain zero, despite all 28 enclosed sample means being negative. No R97 actor or R120 sample result is transferred to these new actors.

Remaining scientific requirement. The 64 independent states are the statistical units; risks, methods, matched hinges and five repeated clocks do not multiply sample size. Population coverage is conditional on the frozen actors and the stipulated ideal entropy model for the 40-bit product-uniform state law. Cross-representation population adoption and population superiority are not established.

Current execution and source route. Table II, Table B.I, and the complete fixed population-inference rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.8 B8. Independent verification stops before fitting

Original advisory request.

> Feature selection, interval-aware fitting, rounding, domain validation, and serialization are not independently reproduced. This is the critical new acquisition stack.

Completed R120 author response. The separately implemented rational validator actually evaluates all 861 unique returned policy keys, with 111,069 terminal contexts and 4,096 complete-policy comparisons. Its true Lagrange affine supports also give an original all-action optimum lower witness, rather than attaching an unproved slope to a scalar lower value. Thus the independent calculation covers regret and action-distance bounds as well as policy value. All 4,096 intervals overlap: 726 production intervals contain the independent enclosure, while 3,370 narrower production intervals are contained in it. Only the former direction inherits coverage from the independent enclosure; overlap or the latter direction does not independently prove fine production endpoints. The validator does not import the production terminal compiler, retains its exact rational traces and 1,806.153-second cost, and claims neither external authorship nor a complete historical fitting-optimizer replication.

The exact completed source/evidence route includes `results120/independent-attempt01/EXECUTION120.json`; the current response map binds all corresponding files.

Completed R121 mathematical, algorithmic and empirical action. Independent verification now begins at the actual finite fitting problem. A separate standard-library rational implementation reconstructs the stored feature values and derivatives, coefficient rounding, objective intervals and primal/dual witnesses for all 56 actors, executing 771,005 assertions. There are 48 actual optimizer fits and eight exactly matched hinge copies, which are not counted as independent fits. For the 28 MARGINAL actors, the finite-box minimax gap is enclosed by exact rational bounds whose largest upper endpoint is approximately 2.58×10^{-8}. The VALUE least-squares audit is reported separately. After fitting, an independent original-primitive evaluator performs 1,406,549 assertions over 167,829 terminal contexts and all delivered and first-proposed policies.

Remaining scientific requirement. The audited problem has fixed features and a prespecified coefficient box. It does not certify globally optimal hidden features, the best unrestricted approximant, the unexecuted global feature cover, or the rounded future controller by exact policy equality. Independence here means a separate implementation, not external human authorship.

Current execution and source route. Proposition 8, Table B.III, and the independent fit and policy execution receipts. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.9 B9. The active submission remains cumulative and disproportionate

Original advisory request.

> A 47-page article, 37-page supplement, and 230-page appendix are excessive for the narrow current theorem and negative acquisition experiment. Historical preservation need not determine the active submission's length.

Completed R120 author response. The active main argument now follows the original continuation/actor problem, the action-dependent modulus, a usable complete-policy improvement rule, the executed matched comparison and its economic interpretation. The supplement contains every new proof and the independently implemented original-optimum construction. The development tables retain all 64 objective/cost cells, 192 clocks, 192 threshold events, 992 sample contrasts and every prescribed price comparison. The 35 original report bodies and the 1,049 inherited item/parent routes remain accessible with exact R119 source bindings. Their presence does not replace the new evidence: the main text reports P's adverse timing, unresolved incremental economic advantage and the stronger conventional proposal alongside its valid gains and the one observed mixed-bound acceptance.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The active account is organized around the original investment decision, integrated-risk marginal learning, its complete mathematical assumptions and the executed economic comparison. The supplement carries complete new proofs, including derivative jumps, boundaries, action-path transport, stationarity and recursive anchors; development material carries the full 58-policy, 28-contrast, clock, cost and price tables. Historical theorems, corrigenda, raw evidence and all 35 original reports remain source-bound. The exact 30-item current response retains every original quotation and every complete R120 map record. The 1,049 inherited item/parent routes remain recoverable without being counted as 1,049 independent or closed scientific issues.

Remaining scientific requirement. Source organization is not evidence that the unresolved neural or economic contribution has been achieved. Final document sizes and source/PDF identities are read from the actual build and publication records. The dated theorem/code registry is a structural index, not a new proof-by-proof revalidation of every historical source.

Current execution and source route. The current source structure, `CONTENT_RETENTION_MAP121.md`, `SOURCE_READY121.json`, and `EVIDENCE_MANIFEST121.json`; later source-specific CI and publication receipts have their own identities. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.10 B10. The negative result is more compelling than the platform claim

Original advisory request.

> The strongest new fact is that a valid global residual objective can fail to improve delivered decisions. The manuscript should either make that misalignment central or supply a new action-aligned objective and theorem.

Completed R120 author response. The new method buys information about an original action contrast and can change the decision. The continuation modulus, fixed-tail information bounds and quantitative tail-defect/refinement analysis are proved; the completed-incumbent gate and the three support proposals retain their R108–R110 ownership. The completed diagnostics distinguish use from independent contribution: the mixed upper is active at 5,961 of 9,016 paid levels and necessary at two accepted levels representing one Poly3 candidate shared by P and D. The transferred incumbent tail-defect bound is inactive in all 2,290 acquired density pairs. P's final action differs from L in 12 cases, with nine exact cell identities and seven unresolved mean differences; its full cost is higher in all 16 primary cells. We claim the proved comparison and actual fixed-resolution decision example, while preserving the unresolved incremental competitiveness and the unfinished action-aware acquisition problem.

The completed evidence for this item is in Table I.III, Table F.II; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The new experiment produces a sharper scientific obstruction inside the NBO problem: accurately solving an action-marginal finite training objective still fails to improve the delivered original decision. All 28 node residuals fall, all 28 sample mean gains are negative, first rejections rise to 294 from 60, and the full primary cohort service is slower in 24 comparisons. At the same time, all 3,712 delivered policies satisfy the prespecified production qualification and independent evaluation records their original regret and action-error bounds. This separates successful finite fitting and a functioning safeguard from the unachieved learning advantage. The action-path and KKT results specify the missing off-design quantities that a stronger method must actually control.

Remaining scientific requirement. These adverse findings do not complete the original ambition of a computationally competitive neural Bellman operator. Global action-path certification, cost-effective precision allocation and a strict equal-information economic advantage remain research obligations. The paper retains its original problem and Econometrica objective while documenting this unmet mechanism.

Current execution and source route. Table II, Table B.II, Table B.IV, and Theorem 9. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.11 M1. Recast the paper around one method-neutral question

Original advisory request.

> Ask when a stored continuation can be safely purchased for recursive-risk dynamic decisions. Treat neural networks as one optional representation, not the source of validity.

Completed R120 author response. Neural Bellman Operators remains the same original-economic-decision author project. The main question is how learned continuations, recursive feasible actors and action-relevant information can produce valid improvements at justified complete cost. The investment model now supplies actual policy changes, original-value/regret

bounds and fixed-law gain inference, rather than an internal fit metric alone. Its continuous shock changes stock imbalance, while investment trades current adjustment cost against the original risk-sensitive continuation and capacity constraint. The main interpretation also confronts the strong online and direct controls and full paid work. The new safeguard is shared across representations; those facts preserve the neural Bellman research identity without assigning an unsupported architecture-exclusive mechanism.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The author retains Neural Bellman Operators and the Econometrica numerical-methods objective. The scientific concentration requested here is implemented within that research identity: can learning the marginal of the integrated original continuation risk improve the feasible original actor after qualification and full information costs? The revised theory and the frozen VALUE/MARGINAL experiment directly address that question. The executed answer is adverse for this design: every paired sample mean favors VALUE, all fixed population intervals remain unresolved, and the ReLU full-menu price sets are empty. Both the shared methodological contribution and the unachieved neural improvement are stated explicitly.

Remaining scientific requirement. Literal redirection to a different paper or venue is not followed. A negative result is not relabeled as completion of the neural economic objective; the remaining action-path, information-efficiency and adoption questions continue inside the original model.

Current execution and source route. The main theoretical sequence and Table II; the unchanged original report and complete prior treatment remain above. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.12 M2. Establish the true novelty of the information theorem

Original advisory request.

> Give a theorem-by-theorem comparison with information-based complexity. Add a genuinely new randomized, nonuniform-law, multivariate, or Bellman-compositional result, or present the current theorem as a specialized supporting lemma.

Completed R120 author response. The novelty statement distinguishes the new original constrained-AVaR smoothness calculation, action-distance integral interface, matching fixed-tail query order and quantitative constrained-tail transfer from the inherited common-risk identity and strict policy gate. The original model's additional smoothness gives its own stronger upper coefficient and is excluded from the generic nonsmooth lower-class claim. The full finite-resolution theorem includes paid deficits and the true future-policy floor. Actual execution shows one Poly3 acceptance at a paid resolution that the separate convex upper cannot authorize at that same resolution. It does not measure a traditional method after another refinement, a counterfactual time saving, or general end-to-end Bellman optimality. The economic and timing comparisons retain those precise limits.

The completed evidence for this item is in Table I.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The revised novelty statement separates known ingredients from the original-risk construction. Action-null quotients, bounded convex readouts, dual witnesses and KKT regret bounds retain their historical and classical attribution. The new proof derives constrained integrated-AVaR marginal regularity from the investment primitives, obtains paid outward probe labels, verifies the finite stored problem, and carries interval marginal errors through nonsmooth feature ranges into feasible action comparisons while preserving recursive anchors. The actual 2,756 original root-Q contexts and separate fit audits make that construction executable. The experiment establishes no optimality of its information complexity and no neural-exclusive representation gain.

Remaining scientific requirement. The inherited deterministic point-value univariate fixed-α lower bound does not become a general Bellman lower bound through this application. Matching bounds at the original-formula interface and a decision-sensitive learning guarantee beyond the stated cover assumptions remain unproved.

Current execution and source route. Theorem 6, Proposition 7, Proposition 8, and Theorem 9. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.13 M3. Develop an action-aligned acquisition objective

Original advisory request.

> Construct a loss that controls rejection, endpoint work, or policy-cost resolution on a predeclared context law. Prove the connection and validate it prospectively against complete work.

Completed R120 author response. All 16 stored model/risk artifacts were freshly acquired under the frozen ordinary original-V_1 interval-midpoint value-label rule, with 256 common sites per risk and complete fitting/audit costs. The new action-relevant acquisition occurs inside the paid policy-comparison interface; it does not change that training objective. Its original-policy consequences are independently evaluated, including 42 positive learned-incumbent gain groups and the unresolved P-versus-L incremental mean comparison. The item therefore separates the completed comparison-information theorem from the still-needed action-aligned loss, acquisition theorem and newly fitted-actor validation. R98's distinct seven-objective acquisition experiment remains preserved under its own unchanged actor and sample barriers.

The completed evidence for this item is in Table F.I, Table F.II; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. An action-aligned acquisition target is now actually implemented: the known current-action cost is kept exact, the critic fits intervals for the original integrated continuation-risk marginal in original action units, and its bounded minimax readout has an independent finite optimality audit. VALUE receives the same

probes and other paid information. This change lowers all 28 node residuals, but the final actions differ in 889 pairs and the independently bounded sample mean gain is negative in all 28 treatments. The action-path theorem explains why a node fit alone cannot provide the needed off-design guarantee; the proof requires a feature-range cover that this campaign has not certified.

Remaining scientific requirement. An implemented marginal target is not yet a successful policy-improvement or acquisition-allocation mechanism. A globally certified path budget, an economically optimal stopping rule and a positive full-cost learning benefit require further algorithmic development and a separate prospectively frozen experiment.

Current execution and source route. Proposition 7, Theorem 9, Table II, and Table B.III. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.14 M4. Expand the conventional geometry frontier

Original advisory request.

> Compare ReLU with conventional oblique hinges, adaptive triangulation, convex regression, and representation-neutral basis selection under equal information, rank, acquisition, and validation budgets.

Completed R120 author response. The contemporary frontier contains online semi-analytic, quadratic, ReLU, equal-feature hinge, spline, degree-two polynomial, degree-three polynomial and grid arms, each with C, L, P and D. All 992 unordered within-risk cell comparisons are retained. Every learned completed cell is strictly worse than online in sample original value, and all 512 ReLU–hinge policy pairs are identical under full equal acquisition charges. D has a strictly better independent mean than L and P in five cells, two exact online identities and nine unresolved comparisons; it is faster than P in every matched clock. Thus conventional geometry, direct original-formula access and the shared certificate all receive their actual competitors. A favorable comparison with only a neural arm's own incumbent is not used as the representation frontier.

The completed evidence for this item is in Table F.I, Table I.VIII; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The executed conventional frontier includes quadratic, degree-two and degree-three polynomial, spline, grid, ordinary hinge with the identical ReLU features and coefficients, and direct online original-formula solution. Their information, feasible actor domain and qualification tolerances are held fixed within the prescribed comparisons. The hinge pays the parent learned acquisition and audit charge and matches every one of the 512 primary ReLU raw and delivered actions. Every one of the 32 primary and 160 individual-clock ReLU full-menu upper not-ruled-out price regions is empty. This treats equivalent geometry as equivalent rather than attributing an ordinary hinge construction exclusively to a network.

Remaining scientific requirement. The fixed feature menus and two acquisition budgets do not exhaust conventional approximation or neural architectures. An exclusive neural claim needs a resource-matched separation beyond these identical-feature controls; the present data supply none.

Current execution and source route. Table B.III, Table B.IV, Table B.VII, and the equal-feature control rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.15 M5. Treat the negative result as first-class evidence

Original advisory request.

> Report aggregate and cell-level rejection changes beside span improvements, explain where the global objective allocates approximation effort, and do not label 36/36 span reduction as delivered-service success.

Completed R120 author response. The main text and complete development tables retain P's 16 primary cost losses and all 48 fresh-clock losses to L, nine exact and seven unresolved P/L mean comparisons, and all six zero-gain online population groups. P retains 2,257 rejected candidates, including 785 exhausted precision ladders; D retains 633 rejections, including 194 exhausted ladders. The inactive transferred-deficit result and the distinction between 5,961 active mixed bounds and only one necessary economic candidate are reported directly. The older R97 residual-span mismatch, adverse R119 phases, conventional wins and unverified neural-adoption regions remain unchanged. No discarded clock, unresolved sign or failed attempt is removed to make the new mechanism appear uniformly favorable.

The completed evidence for this item is in Table I.III, Table F.II; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. Every primary raw proposal, qualification attempt, rejection, returned policy and matched timing row is retained. The 4,066 attempts include all 354 first rejections and their conventional fallback; none is removed from costs or independent evaluation. The new adverse pattern—28 improved node fits with 28 worse sample means, 294 versus 60 first rejections, and no ReLU full-menu price region—is presented alongside all favorable state-level threshold events. The 28 mean comparisons are not independent replications, since the same 64 states and matched feature controls recur. R49, R97, R119 and R120 negative evidence remains unchanged and is not used in place of this new execution.

Remaining scientific requirement. No adverse cell, unfavorable clock or failed historical attempt is converted into a favorable aggregate by reselection or retiming. Current sample signs do not establish population harm: all 28 fixed population mean-gain intervals still contain zero.

Current execution and source route. Table II, Table B.II, Table B.V, and the complete signed threshold rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.16 M6. Execute a consequential application or narrow the claim

Original advisory request.

Either calibrate policy margins, compute prices, fees, and workloads in an application where adoption matters, or state that the paper is about numerical certification rather than economic purchase.

Completed R120 author response. The economic interpretation reports the original risk-sensitive investment objective, not a different loss or welfare surrogate. It shows how the independently bounded ReLU own-incumbent gains reduce original regret and action-distance ceilings, and why a larger spline gain can reflect a worse incumbent. The online semi-analytic policy is already strictly better than every learned completed policy on the sample, while direct refinement is a material traditional competitor. The simultaneous 42 positive learned-incumbent lower bounds and all actual price scenarios are reported with their population and cost scopes. The units remain normalized original-objective units: this two-date experiment adds decision-relevant evidence but does not claim monetary, consumption-equivalent or field-adoption calibration.

The completed evidence for this item is in Table F.I, Table F.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The actual application remains the original two-date constrained investment problem with two state coordinates, continuous innovations, nested AVaR at $\alpha = 1/2, 3/4$, and the unchanged implemented future-policy allowance. Independent evaluation now reports original objective intervals, original-optimum support bounds, policy regret, action-error upper bounds and complete computation costs for all 58 cells. All 3,712 delivered policies meet the frozen production qualification, but MARGINAL worsens every paired sample mean and yields no ReLU full-menu adoption region. These are economically interpretable investment decisions under the original objective; they are not gains from a substitute posted-price or recursive-utility model.

Remaining scientific requirement. A consequential externally calibrated application and a demonstrable full-cost neural economic benefit remain unfinished. The preserved stochastic-control, nonlinear recursive-utility and NDU/FBSDE connections retain their own hypotheses; this two-date execution neither validates those different models nor changes the intended research direction.

Current execution and source route. The original-economy definitions, Table B.I, Table B.II, and Table B.IV. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.17 M7. Add a genuine scaling experiment

Original advisory request.

Vary horizon, controls, innovation dimension, and risk level under matched comparators. Report product-point counts, memory, endpoint work, acquisition, and complete service work.

Completed R120 author response. The completed experiment is explicitly two-dimensional, two-date and evaluated at the two prescribed risk levels. It reports each service's full acquisition, original-value queries, candidate/refinement counters, raw process costs and complete resource-vector frontier. The four predeclared amortization volumes are conditions computed from these retained measurements; they are not measured throughput, higher-dimensional scaling or a horizon experiment. The new information theorem has a scalar-tail scope, and the generic lower bound excludes richer original-formula access. Existing higher-dimensional and application evidence retains its own source and limitations. Broader scaling and consequential application claims remain concrete research obligations rather than being filled by these 64 new initial states.

The completed evidence for this item is in Table F.II, Table I.VIII; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. Both prespecified budgets were acquired and fitted in full before the held-out states: 325 and 1,053 original root-Q contexts per risk, a ratio of 81/25. The grid and spline feature counts and all actual acquisition, audit and service costs are reported at both budgets. Among the 28 coarse-to-fine original sample mean comparisons, 15 enclosed gains are strictly positive, 12 strictly negative and one contains zero. These are descriptive comparisons on the common fresh sample, not an added population inference family. Different design-node sets also prevent a finite-node residual ratio from serving as an off-design convergence certificate.

Remaining scientific requirement. The experiment is genuine information-budget scaling at fixed dimension and horizon. It does not establish dimension scaling, horizon scaling, an asymptotic rate or monotone improvement with more data; those remain separate scientific requirements.

Current execution and source route. Table B.VI, Table B.III, and both actual acquisition-process receipts. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.18 M8. Strengthen timing evidence

Original advisory request.

> Use more repetitions, record or control processor state, and make operation counts primary. Predeclare the inferential target for deployment clocks if a probabilistic purchase claim is desired.

Completed R120 author response. All three 32-state timing processes completed, retaining 6,144 timed actions and 192 cohort clocks. Every timed action, interval and recorded query count matches its corresponding primary rule; no mismatch or clock was censored. P is slower than L in all 48 matched clocks; D is faster than P in all 48 and faster than L in 41, with seven losses and two adverse cell medians. Cohort measurements, standalone acquisition charges, loading and larger process/launcher scopes are labeled separately. These are descriptive engineering observations under the observed execution conditions. The clocks are not extra population observations and do not supply confidence bounds for future hardware, latency, energy prices or external load.

The completed evidence for this item is in Table F.II; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. Five fresh timing processes execute the fixed first 32 states in the same catalogue, and all 9,280 saved timing actions match the primary state and action identities. Among 140 matched MARGINAL/VALUE cohort clocks, MARGINAL is faster in 40 and slower in 100; including the measured allocated acquisition and audit bill changes these counts to 38 and 102. In the 64-state primary cohort it is faster in four of 28 complete-service comparisons. All five clocks, process loads, serialization, rejected proposals and fallback are retained, and the selected study-process ledger includes the independent policy validation. Nested timing processes are not added again to the enclosing primary clock.

Remaining scientific requirement. Five descriptive observations are not a runtime confidence interval or a claim about the machine-independent timing distribution. Allocated standalone deployment costs are calculated from measured components and are distinguished from directly observed whole processes. Repeats do not add independent economic states.

Current execution and source route. Table B.IV, Table B.V, and the selected research-stage cost rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.19 M9. Independently reimplement fitting and package a standalone environment

Original advisory request.

> Reproduce feature selection, interval-aware fitting, rounding, whole-domain validation, and serialization independently. Include a container or lockfile and all inherited modules needed to run the new tests from the artifact alone.

Completed R120 author response. The exact source F, actual actor M, new sample D and pre-outcome reporting specification R are preserved together with their real launch and publication receipts. Primary evidence was separately published as E0 before final reporting. Independent original-primitives validation, the fixed rational statistics and the first read-only reporter have all completed with exit zero; the reporter binds 777 unchanged input files and 36 stable products. The derivative algebra diagnostic preserves its first source-path failure and the corrected second attempt, without changing frozen science. The bounded package audit and source CI distinguish record/byte checks and statistical/report replays from a fresh primitive-economy rerun. Actual final source, PDFs, build logs and CI states belong to the separate source/PDF and publication declarations; no R119 success or future R120 job is asserted here as a completed R120 CI.

The exact completed source/evidence route includes `provenance/SOURCE_PUBLICATION120.json`; the current response map binds all corresponding files.

Completed R121 mathematical, algorithmic and empirical action. The independent fitting implementation has now executed on all 56 stored actors, checking exact feature and derivative assembly and finite primal/dual witnesses without calling the fitting solver. The separate original-primitive evaluator has completed all 1,301 distinct raw and delivered policy keys with 1,406,549 assertions. The fixed inference calculation and deterministic reporter also exited successfully. The reporter binds 473 input files, checks 220,310 record assertions and reproduces 49 stable output files without fitting, new states, original economic evaluations or timing reruns. The standalone

source, dependency pins, immutable inputs and reproduction commands distinguish this bounded reporting check from a full scientific replay.

Remaining scientific requirement. A separate implementation is not external authorship, and the finite fitting certificate does not optimize hidden features or establish a global feature cover. Final source-specific CI and publication evidence are distinct from the already successful scientific stages and from the M/R transport workflows; they are identified by their own receipts without a future-success assertion inside this source.

Current execution and source route. The independent fit/policy receipts, `SOURCE_READY121.json`, `EVIDENCE_MANIFEST121.json`, and the source-specific CI/publication records. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.20 M10. Shorten the active submission substantially

Original advisory request.

> Move historical material to an archival companion. The active article should contain only the current qualification theorem, precise information result, acquisition objective, matched experiment, and supported economic interpretation.

Completed R120 author response. The main paper concentrates on the original recursive decision, complete new mathematical argument, contemporaneous experiment and economic interpretation. Full assumptions, proofs, independent optimum verification and all detailed tables remain in the supplement/development companion. All 35 exact original reports are copied unchanged into the current source package; the 620 major, 394 additional and 35 overlapping-parent treatments retain their original identities and specific remaining requirements. The current response supplies an actual proof/evidence route for each of the 30 controlling items. This organization preserves NBO's learned continuation and actor problem, the strong traditional comparisons and every valid historical result, while keeping the current contribution assessable without treating document reorganization as scientific closure.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The principal narrative now follows one economic problem, the original-risk marginal acquisition and fitting construction, its action-path and stationarity theorems, and the completed 28-treatment economic test. Full proofs reside in the supplement; all policy, fitting, budget, timing and price tables and the research history have explicit development routes. The complete original advisory quotations and R120 treatments remain in this response and its machine map. Historical proof and code sources, including the original corrigenda and negative results, remain recoverable by commit and hash rather than being silently deleted to shorten the article.

Remaining scientific requirement. Final page counts are facts of the actual strict build, recorded in the retention and publication records. Structural reorganization does not close the unresolved contribution questions, and a historical syntax/index scan is not a new correctness proof.

Current execution and source route. The main/supplement/development wrappers, `CONTENT_RETENTION_MAP121.md`, `SOURCE_READY121.json`, and the actual source/PDF publication records. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.21 Minor 1. Error-versus-information terminology

Original advisory request.

> 1. State 'N^{-2}' consistently as an error-versus-information rate; "information exponent two" is easy to misread.

Completed R120 author response. The new result is called interval width versus scalar point information, with order $\min\{KR(1-\alpha)N^{-2}, ER(1-\alpha)N^{-1}\}$. Original-policy contrast, certified gain, regret, value width, squared action-distance bound and complete work are separate quantities throughout the proofs and all 64 objective/cost cells. The independently computed optimum is an enclosure, so the displayed RMS quantities are upper bounds rather than measured errors to a known exact optimizer. The achieved R119 N^{-2} tail-uniform AVaR width result retains its established attribution; the executed economic gains require their own policy evidence.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The inherited N^{-2} statement remains an error-versus-information bound under its deterministic scalar oracle convention. The new marginal construction displays its finite probe and label tolerances separately from fitted node error, feature-cover error and action regret. The actual budgets contain 325 and 1,053 original root-Q contexts per risk, and the 28 coarse-to-fine sample comparisons include both positive and negative gains. These two budget observations are reported as executed information scaling, not evidence for an asymptotic exponent or a universal neural convergence rate.

Remaining scientific requirement. No rate for general Bellman information complexity, end-to-end time or economic gain is inferred from a function-class approximation rate or from the two executed budgets.

Current execution and source route. Proposition 7, Theorem 9, and Table B.VI. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.22 Minor 2. Oracle qualifiers

Original advisory request.

> 2. Put "deterministic, point-value, univariate, fixed-alpha" adjacent to every high-level summary of the lower bound.

Completed R120 author response. Deterministic, exact scalar point-value, univariate, uniform-law, convex-Lipschitz and fixed-known-tail qualifiers appear with the new information statement. Purchasing the two continuation values at a point costs two queries. The original smooth investment formula is a narrower class than the lower adversary, and risk-tail localization, finite point precision, terminal solves and the future controller remain separately paid. R119's risk-level-uniform width theorem is retained under its own assumptions. The completed two-risk experiment does not establish randomized, multivariate or arbitrary-law oracle optimality.

The completed evidence for this item is in Table I.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. Every operative interpretation of the inherited lower bound retains the deterministic, point-value, univariate, fixed-α oracle restrictions and its function class. R121 supplies a constructive original-model acquisition interface and its actual paid queries, not a broader lower-bound proof. The direct original-formula online control is retained in all economic comparisons; excluding it on the basis of the narrow oracle theorem would give an unfair computational claim.

Remaining scientific requirement. Randomized, multivariate, nonuniform and formula-oracle lower bounds remain separate open statements. The actual original-model information bill does not itself establish minimax optimality.

Current execution and source route. Theorem 6, Proposition 7, and the acquisition and complete-frontier rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.23 Minor 3. Tail dependence

Original advisory request.

> 3. Discuss the opposite tail-level behavior of the lower and upper constants as 'alpha' approaches one.

Completed R120 author response. The achieved R119 order $KR(1-\alpha)N^{-2}$ is preserved. R120's fixed-tail comparison intersects widths bounded by $2KR(1-\alpha)/n^2$ and $ER(1-\alpha)/n$, with the original smooth coefficient supplied separately. The action modulus includes its explicit $\alpha/(1-\alpha)$ dependence; this prevents a small tail mass alone from implying a cheap original-model computation near $\alpha = 1$. The full-support $\alpha = 0$ boundary is separate and the normalized-tail statement excludes $\alpha = 1$. Actual new observations use only the predeclared $1/2$ and $3/4$ risk levels and retain every paid tail/terminal operation.

The completed evidence for this item is in Table F.II, Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The original-risk regularity and probe propositions retain explicit dependence on the tail probability and capacity constants, while the inherited fixed-tail lower bound keeps its own tail-dependent constant and function class. The executed comparison reports both $\alpha = 1/2$ and $\alpha = 3/4$ separately throughout the acquisition,

regret, loss and cost tables; the 28 paired sample mean gains are negative across both risk levels. No data-dependent risk level has been chosen to improve the claim.

Remaining scientific requirement. The two fixed risk levels do not provide empirical evidence about the $\alpha \uparrow 1$ limit. Opposite movements of constants from different bounds or function classes cannot be combined into a uniform sharp-rate claim without the corresponding assumptions and proof.

Current execution and source route. Theorem 6, Proposition 7, and Table II. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.24 Minor 4. Acquisition restrictions

Original advisory request.

> 4. The readout coefficient bounds are acquisition restrictions, not characterizations of the best approximation; report sensitivity.

Completed R120 author response. The actual actor artifacts and acquisition table preserve all prescribed feature counts, coefficient signs/bounds, rounding and ties. The constrained rank-six readouts are compared with the larger fixed polynomial and grid controls under the same 256 original-V_1 label sites per risk. ReLU and ordinary hinge have identical retained parameters and actions, with the full shared-fit charge assigned to each standalone deployment. These are acquired-model comparisons under known restrictions. They are not best-approximation results for unrestricted representation classes, and the experiment supplies no post-outcome coefficient or architecture sensitivity menu.

The completed evidence for this item is in Table F.I, Table F.II; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The actual fitting problem fixes a coefficient box of 128 and 30-bit coefficient storage; its exact audit certifies this finite feasible class. All 28 MARGINAL primal/dual gap upper bounds are at most approximately 2.58×10^{-8}, and VALUE least-squares gaps are separately reported for the VALUE actors. The two acquired budgets change information and some feature counts, but the coefficient box is unchanged. We therefore report the restriction and its exact solved-problem gap without interpreting it as the best unrestricted approximation or as a coefficient-box sensitivity experiment.

Remaining scientific requirement. Sensitivity to the coefficient bound has not been executed in R121. Changing that bound, features or selection after seeing these outcomes would require a new declared experiment; the present two-budget comparison cannot answer the omitted box-sensitivity request.

Current execution and source route. Proposition 8, Table B.III, and Table B.VI. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.25 Minor 5. Residual span and action gap

Original advisory request.

> 5. Distinguish validated residual span from Bellman action gap in terminology and tables.

Completed R120 author response. Every primary group now has separate original-policy bounds, independent original-optimum bounds, mean regret, RMS action-error upper bound and own-incumbent gain, alongside production contrast and certificate counters. The independent validator supplies 55,104 retained all-action support rows and all 4,096 policy comparisons. For example, the ReLU P gain is economically positive against its incumbent while its mean difference from L is exactly zero. A narrower value interval, smaller fit residual or lower regret ceiling is therefore never counted as an observed policy-value decrease by itself. The 726 versus 3,370 containment directions also retain their different independent-verification implications.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The result tables distinguish finite-node marginal residual, finite-box optimization gap, production qualification, independently bounded original-policy regret and action-error upper bounds. The distinction has empirical content: all 28 node residuals decrease while all 28 original sample mean gains are negative. All 3,712 policies pass production qualification, yet that does not establish a better learned decision than VALUE or a neural adoption advantage. Production and independent interval overlap is described only as compatibility; it does not independently certify the narrower production endpoints.

Remaining scientific requirement. A node residual or fitting gap is not an off-design Bellman action gap. The proved feature-cover and KKT implications require their stated hypotheses; the fitted global cover was not numerically certified, and action-error upper bounds are not exact observed action errors.

Current execution and source route. Theorem 9, Proposition 10, Table II, and Table B.I. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.26 Minor 6. Rejections and unsuccessful work

Original advisory request.

> 6. Report rejection changes beside every span improvement.

Completed R120 author response. Every attempted candidate and resolution, rejection, exhausted ladder, unchanged action and fallback is retained in the actual traces and complete tables. P's 2,257 refusals and D's 633 refusals appear beside accepted-step counts, as do 785 and 194 exhausted ladders. Mixed-bound use is reported at all 9,016 paid levels: 5,961 active upper bounds and two necessary accepted records for one economic candidate. The transferred incumbent bound is inactive in all 2,290 density pairs. Its mathematical validity is not converted into an

unobserved reduction of already purchased tail endpoints, candidate-upper work or total time. Every unresolved and adverse comparison remains included.

The completed evidence for this item is in Table I.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. Each of the 28 loss-contrast rows reports the node-residual change beside the first-rejection change and the independently bounded original gain. Although every node residual improves, rejection counts rise in 18 paired treatments, fall in two and are unchanged in eight; the aggregate is 294 MARGINAL rejections versus 60 VALUE rejections. All 354 rejected first proposals are kept, independently evaluated and charged together with fallback. The raw-proposal table and delivered-policy table distinguish a poor learned suggestion from the corrected policy ultimately returned.

Remaining scientific requirement. A smaller residual is not evidence of fewer rejections or safer raw actions. The 28 treatments reuse the same states and include identical-feature controls, so these counts are descriptive treatment totals rather than 28 independent draws.

Current execution and source route. Table II, Table B.II, and the complete qualification-attempt rows. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.27 Minor 7. Observed engineering ranges

Original advisory request.

> 7. Label the acquisition bracket as an observed engineering range, not a confidence interval.

Completed R120 author response. The three process-specific measurements, all 192 cohort clocks and their medians and ranges are labeled descriptive engineering observations. All 6,144 comparisons of actions, intervals and query counts match, so the reported conditional pairing of timing and economic outcomes is actually checked. These clocks remain distinct from outward rational value enclosures and from the explicitly proved simultaneous 95% population family of 48 groups and 192 events. Timing replicates do not increase its 64 independent initial-state units. The inherited seven R119 clocks retain their separate episode and do not enlarge the current sample.

The completed evidence for this item is in Table F.II, Table F.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The acquisition ledger records the two actual processes and the component allocation used for deployment comparisons; it does not assign sampling coverage to the observed engineering costs. The five fresh service clocks are all reported, with 40 faster and 100 slower MARGINAL cohort comparisons, or 38 and 102 after the allocated acquisition and audit bill. Raw clock observations, complete enclosing processes and calculated reuse costs have separate fields. The independent original-policy validation cost remains visible in the research ledger.

Remaining scientific requirement. Observed minima, maxima, medians and component allocations are not runtime confidence intervals. The fixed 95% family concerns economic state sampling, not runtime variation; timing repeats do not enlarge its sample.

Current execution and source route. Table B.V, Table B.IV, and the acquisition/research process records. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.28 Minor 8. Adoption over the conventional frontier

Original advisory request.

8. State visibly that there are zero adoption regions over quadratic least squares.

Completed R120 author response. The old zero-adoption-over-quadratic result and the R119 conventional frontier remain explicit. In the new fixed comparison, positive gains against learned incumbents coexist with an online advantage against all 14 learned completed means and exact ReLU–hinge identity. All 32 ReLU intersections against the predeclared full conventional menu, across four amortization volumes and three clocks, have empty sufficient-price sets and empty intersections of upper not-ruled-out sets. Among individual clock comparisons for ReLU, 12 of 96 upper regions remain nonempty; two of eight actual-primary workload upper regions are also nonempty. These are unresolved comparisons, retained separately from the three-clock intersections. The 17 nonempty sufficient menu regions across representations belong to completed conventional policies. The 112 separate actual-64-state price domains retain the actual workload and charges. These strict recorded-menu conditions and own-incumbent gains do not assert population adoption or rule on every possible deployment criterion.

The completed evidence for this item is in Table I.VIII, Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The original R97 zero certified adoption result over quadratic least squares and the exact R120 individual-clock qualifications remain explicitly preserved above. The completed R121 full-menu result is now stronger for its own declared scenarios: zero of 32 primary ReLU price cells, zero of 160 individual timing cells and zero of 32 all-five-clock intersections has a nonempty sufficient or upper not-ruled-out region. The predeclared full menu includes quadratic least squares and the other conventional alternatives at both budgets. Every pair constraint remains in the complete table, so the full-menu result is not silently relabeled as a separate pairwise claim against quadratic alone.

Remaining scientific requirement. An empty full-menu region does not logically imply that every pairwise comparison is impossible, and the older unresolved R120 individual-clock regions remain unchanged. These R121 finite-cohort, measured-cost conditions are not population adoption or a universal impossibility theorem.

Current execution and source route. Table B.VII and every exact pair constraint, menu and all-five-clock intersection. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.29 Minor 9. Verified approximation and HJB literature

Original advisory request.

> 9. Compare the certification contribution carefully with existing verified approximation and HJB-residual literatures.

Completed R120 author response. The main text locates the mixed integration interface relative to retained verified approximation, Bellman/HJB residual ideas, constrained safeguards and the source-owned common-risk identity. Complete proofs retain the original filtration, boundaries and one-sided future-policy allowance. Exact binomial inversion and bounded-mean concentration are attributed as classical ingredients; the new population conclusion is the specified transfer from actual certified gains. The stopped Brownian verification subcase retains its hypotheses and complete boundary argument. The present two-date validation supplies no proof or experimental substitute for a general coupled NDU/FBSDE, nonlinear recursive-preference application, temporal-self equilibrium or game.

The completed evidence for this item is in Table F.III; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. The certification account retains the verified approximation, Bellman/HJB residual, recursive-utility and safeguarded-policy connections with their original hypotheses and attribution. R121 adds a complete original-risk marginal construction through constrained tails, a paid interval-label interface, a finite fitting witness and an explicit nonsmooth feature-range transfer. The independently implemented finite audit and original-policy evaluator have both executed, while the global fitted-feature cover remains unexecuted. Classical concentration and binomial inversion ingredients do not become new neural theorems because they support the fixed R121 evidence.

Remaining scientific requirement. The preserved NDU/FBSDE, nonlinear preference, temporal-self and game links require their own well-posedness, filtration, boundary and equilibrium assumptions. The present two-date investment validation neither proves nor executes those different economic claims.

Current execution and source route. The complete new proofs, retained correctness/bridge sources, Table B.III, and the independent original-policy receipts. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

2.30 Minor 10. Title and neural content

Original advisory request.

> 10. The title should not carry more neural-specific content than the theorem and evidence identify.

Completed R120 author response. The title and active main text retain Neural Bellman Operators as the study of learned original continuations, recursive operators and feasible actor/critic decisions. The new action-dependent comparison and safeguard are attributed to an eligible-method interface shared by the neural and conventional representations. Their original-policy gains are actually measured and independently bounded; conventional hinge remains identical to ReLU, and the strongest original-formula controls remain visible. This establishes a concrete shared numerical-methods result inside the NBO problem while leaving equal-information neural advantage and action-aware learning as explicit ongoing research requirements.

The completed evidence for this item is in Table F.I; the exact row files and hashes are retained in the current response map.

Completed R121 mathematical, algorithmic and empirical action. Neural Bellman Operators remains the research identity: recursive continuation operators, their learned critics and feasible actors are studied under the original investment objective. The title and contribution distinguish the neural implementation from the representation-neutral theory. Current evidence is stated directly: the marginal objective is accurately solved at the finite nodes, yet every paired sample mean worsens, matched hinge actions equal ReLU, and no ReLU full-menu adoption region is obtained. These findings inform the next original-model algorithmic problem rather than redefining a shared hinge geometry as neural-specific success.

Remaining scientific requirement. The title does not assert an established neural-exclusive economic or information advantage. Achieving such an advantage with equal information, feasible policies and complete costs remains part of the continuing NBO and Econometrica objective.

Current execution and source route. The revised main contribution statement, Table II, and Table B.VII. The item-specific exact table hashes and successful current-stage receipts are bound in `provenance121/R97_REQUIREMENTS121.json`. These are completed R121 observations; the retained R120 evidence above has its own source and sample.

3 Historical preservation, unresolved episodes and publication scope

All earlier original report text and author response routes remain unchanged in the compact inherited maps indexed by `provenance121/reports121/REPORT_SOURCE_INDEX121.json`. The 620 major routes, 68 explicit minor requests, 295 separately retained supplementary requirements, 31 embedded requests and 35 overlapping parent treatments total 1,049 authored route records. The overlapping parent treatments are not added as independent scientific issues. Exact source texts and their original commit identities remain authoritative.

R98 and R114/R115 retain their independent protocols, selected-model and sample barriers. A previously recorded not-started state is a dated observation, not a live process-status assertion. R111–R118 retain their separate model and information conventions, incomplete handoffs, failed transport and actual later reproductions where documented. In particular, R119's actual unchanged R117/R118 reproductions do not complete the independent R115 population study. R118's inherited R110 article and truncated transport do not constitute a newer completed author baseline. The retention map records these episode-specific boundaries: `CONTENT_RETENTION_MAP121.md`.

The original-theory and source registries distinguish a complete file or cross-reference census from full semantic reading and from fresh mathematical revalidation. Corrected historical claims

remain visible with their corrigenda; their original source is not silently edited. The report reading-coverage receipt records the actual reader and scope of each full report reading rather than inferring reading from a hash.

R120 source CI run 39240515942 actually completed successfully at its bound source and produced artifact 12181820272. Its independently read artifact and exact four-PDF comparison are retained as R120 evidence. They are not R121 CI. The final R121 source, PDF and publication declarations will be separate later records bound to their actual outputs; a queued run, a successful archive copy or a bounded reporting replay will not be described as a complete scientific rerun.